DROPZYY 1 - COMPREHENSIVE QA TESTER VERIFICATION SUITE

Target Release: v2.4.0 (Production RC)
Generated: October 26, 2026

Official End-to-End Master Checklist for Functional, Security (Zero-Leak Auth), Hero Banners, Cart/Checkout, Mobile UI & Admin Operations

Host: freshkart-9j9b.onrender.com
Databases: MongoDB Atlas + SQLite

Lead QA Tester:	_____	**Execution Date:**	_____	**Scope:**	**42 End-to-End Test Cases**
Devices Tested:	Mobile (iOS/Android), Desktop (1080p+)	**Browsers Tested:**	Chrome, Safari, Edge, Firefox	**Overall Verdict:**	[ ] PASS [ ] FAIL [ ] CONDITIONAL

EXECUTION GUIDELINES & SEVERITY LEGEND: **P1 (Critical):** Blocker / Security breach / Broken core purchase flow | **P2 (High):** Major UI/Functional defect / Inaccurate calculation | **P3 (Medium/Low):** Cosmetic / Non-blocking edge case | **Status Keys: P** = Pass, **F** = Fail, **B** = Blocked

1. USER AUTHENTICATION & ZERO-TRUST SESSION SECURITY (FLIPKART / AMAZON ARCHITECTURE)

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-SEC-01	**User Registration & Validation**	1. Click Login/Sign Up. 2. Enter username, email, phone, password. 3. Submit form.	Account created in MongoDB/SQLite; success toast shown; duplicate username/email rejected with clear error.	P1	P / F / B
[ ]	TC-SEC-02	**User Login & Clean LocalStorage**	1. Enter valid credentials. 2. Click Login. 3. Inspect DevTools -> Application -> Local Storage.	User logs in; localStorage contains strictly {'username': '...'} and NO passwords, roles, or raw tokens.	P1	P / F / B
[ ]	TC-SEC-03	**SessionStorage Zero-Leakage (Flipkart Model)**	1. Open DevTools -> Application -> Session Storage. 2. Inspect all key-value pairs after login.	Zero profile or role data in SessionStorage. User role is never exposed or trusted in client storage.	P1	P / F / B
[ ]	TC-SEC-04	**Role Spoofing Tamper Resistance**	1. Open Console. 2. Execute localStorage.setItem('role','admin') or sessionStorage.setItem('role','admin'). 3. Reload.	Tampered client keys are ignored. User remains standard customer; admin panel and admin routes remain locked.	P1	P / F / B
[ ]	TC-SEC-05	**Server Auth Endpoint Guard (/api/auth/me)**	1. Send GET request to /api/auth/me without valid cookie/bearer token or with forged payload in Network tab.	Server strictly rejects forged requests with 401 Unauthorized; only verified session returns authentic role.	P1	P / F / B
[ ]	TC-SEC-06	**User Logout & Session Destruction**	1. Click Profile Menu -> Logout. 2. Attempt navigating back or invoking protected endpoints.	Session cleared from memory; UI resets to guest state; subsequent protected API calls return 401 Unauthorized.	P2	P / F / B

2. HERO BANNERS, FESTIVE POSTERS & LOCATION SELECTOR

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-BAN-01	**Live MongoDB Banner Fetching**	1. Open homepage. 2. Inspect Network tab for GET /api/banners. 3. Check loaded banner cards.	Banners load dynamically from MongoDB Atlas 'banner_posters' collection with fallback to default festive slides.	P1	P / F / B
[ ]	TC-BAN-02	**Rangoli Festive Poster Display**	1. Observe carousel when Rangoli poster is active. 2. Verify image clarity, title, and badge text.	uploaded-rangoli.png renders crisp & responsive; title 'Festive Rangoli & Pooja Specials' and badge display cleanly.	P2	P / F / B
[ ]	TC-BAN-03	**Action Button Optional Toggle (OFF)**	1. Ensure showButton: false for Rangoli poster. 2. Observe banner card on desktop and mobile view.	Action button ('Shop Rangoli') is strictly hidden (display: none !important) without disrupting card layout.	P1	P / F / B
[ ]	TC-BAN-04	**Action Button Optional Toggle (ON)**	1. Set showButton: true with custom btnText ('Order Now') on a promo slide. 2. View banner.	Action button displays with custom label; clicking button filters products or redirects to specified target category.	P2	P / F / B
[ ]	TC-BAN-05	**Banner Auto-Rotation Timing (7 Seconds)**	1. Load homepage without clicking carousel. 2. Use stopwatch to measure slide transition interval.	Carousel auto-rotates smoothly every 7 seconds (not rushed 3-4s); gives customers ample time to view offers.	P2	P / F / B
[ ]	TC-BAN-06	**Manual Carousel Navigation & Touch Swipe**	1. Click Next/Prev chevron arrows on desktop. 2. Swipe left/right on touch screen or mobile viewport.	Slide switches immediately to next/previous poster; auto-rotation pauses on user interaction and resumes smoothly.	P2	P / F / B
[ ]	TC-BAN-07	**Online / Offline Poster Filter**	1. In Admin panel, toggle poster status to 'Offline'. 2. Refresh customer homepage.	Offline poster is completely excluded from client carousel; only 'Online' active posters are displayed.	P2	P / F / B
[ ]	TC-BAN-08	**Location Selector Outside-Click Dismissal**	1. Click Location Picker in navbar. 2. Click outside dropdown on background page area.	Location selector dropdown closes immediately upon clicking outside anywhere on the page.	P2	P / F / B

3. PRODUCT CATALOG, CATEGORIES & SEARCH FILTERING

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-CAT-01	**Category Bar Navigation (Desktop & Mobile)**	1. Click each category tab (All, Sweets, Groceries, Pooja Essentials, Beverages, Snacks).	Product grid filters instantly to match chosen category; active category tab shows distinct active pill highlight.	P1	P / F / B
[ ]	TC-CAT-02	**Mobile Category Bar Horizontal Scrolling**	1. Switch to mobile viewport (iPhone / Android). 2. Swipe category bar horizontally.	All categories are fully visible and horizontally scrollable without overflowing or clipping on small screens.	P2	P / F / B
[ ]	TC-CAT-03	**Instant Search Bar & Live Filtering**	1. Type search query in navbar search (e.g., 'Rangoli', 'Milk', 'Laddoo'). 2. Clear query.	Product list filters in real time with smooth debouncing; clearing search restores full catalog instantly.	P1	P / F / B
[ ]	TC-CAT-04	**Search Zero-State Handling**	1. Type non-existent query (e.g., 'xyzabc999'). 2. Verify page feedback.	Clear 'No products found' friendly empty state shown with a quick 'View All Products' reset button.	P3	P / F / B
[ ]	TC-CAT-05	**Product Card Content & Pricing Integrity**	1. Verify product card details: Image, Title, Weight/Unit, Discount Tag, Strike-through MRP, Offer Price.	All images load cleanly without broken icons; math (MRP - Discount = Selling Price) is accurate across all cards.	P2	P / F / B
[ ]	TC-CAT-06	**Out of Stock State & Add Restriction**	1. View product with stock = 0 in database. 2. Attempt to add item to cart.	Card displays 'Out of Stock' badge; 'Add' button is disabled; user cannot add zero-stock items to cart.	P2	P / F / B

4. SHOPPING CART, DYNAMIC PRICING & CHECKOUT WORKFLOW

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-CRT-01	**Add to Cart & Quantity Selector (+ / -)**	1. Click 'ADD' on product. 2. Verify button converts to [- 1 +]. 3. Click [+] to 3. 4. Click [-] down to 0.	Cart count increases; price updates dynamically; decreasing to 0 reverts button back to 'ADD' and removes item.	P1	P / F / B
[ ]	TC-CRT-02	**Cart Modal & Floating Mobile Bar**	1. Add multiple items on desktop. 2. Check mobile view for floating bottom cart strip.	Mobile bottom bar displays total items count & grand total; clicking bar opens cart drawer smoothly.	P1	P / F / B
[ ]	TC-CRT-03	**Price Calculation & Delivery Fee**	1. Review cart bill summary: Item Total, Handling Charge, Delivery Fee (Free over threshold), Grand Total.	Sum is mathematically exact; discounts and coupon codes calculate and deduct correctly before checkout.	P1	P / F / B
[ ]	TC-CRT-04	**Checkout Form & Input Validation**	1. Click 'Proceed to Checkout'. 2. Test empty fields, invalid phone (<10 digits), and missing delivery address.	Form prevents submission with clear validation errors highlighting required fields.	P1	P / F / B
[ ]	TC-CRT-05	**Payment Method Selection (COD / Online)**	1. Select Cash on Delivery (COD). 2. Select Online UPI / Card. 3. Place Order.	Selected payment method is recorded with order; checkout proceeds to order confirmation without errors.	P1	P / F / B
[ ]	TC-CRT-06	**Order Placement & Confirmation Screen**	1. Submit valid checkout order. 2. Verify confirmation dialog.	Order ID generated (e.g., #ORD-XXXX); cart resets to 0; sound/visual success animation displayed.	P1	P / F / B

5. LIVE ORDER TRACKING & CUSTOMER MANAGEMENT

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-TRK-01	**Live Order Status Progression**	1. Place order. 2. Open Order Tracker. 3. Transition order status in admin backend (Placed -> Packing -> Delivered).	Tracker stepper advances dynamically with timestamps: Order Placed -> Order Accepted -> Packing -> Out for Delivery -> Delivered.	P1	P / F / B
[ ]	TC-TRK-02	**Estimated Delivery Countdown Timer**	1. Observe delivery countdown timer on active order tracking screen.	Timer counts down accurately (e.g., 15 mins instant delivery promise) with dynamic delivery partner details.	P2	P / F / B
[ ]	TC-TRK-03	**Order History in Customer Profile**	1. Navigate to Profile -> My Orders. 2. Inspect order listing.	Previous orders listed in reverse chronological order with date, item summary, price, and status badge.	P2	P / F / B
[ ]	TC-TRK-04	**Invoice PDF Generation & Download**	1. Click 'Download Invoice' on completed order card.	PDF invoice downloads cleanly with company branding, tax breakdown, delivery address, and ordered items.	P2	P / F / B

6. ADMIN DASHBOARD, LIVE POSTER TOGGLE & STORE OPERATIONS

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-ADM-01	**Admin Route Guard & Privilege Check**	1. Attempt to open Admin Panel as guest or standard customer.	Access strictly denied with 403 Forbidden or redirected to login; admin UI is hidden from non-admin accounts.	P1	P / F / B
[ ]	TC-ADM-02	**Mobile Admin Panel View & Tool Visibility**	1. Log in as admin on mobile device / narrow viewport (<600px). 2. Check admin navigation toolbar.	All admin tools ('Track Bookings', 'Data Analytics', 'Add Item', 'Manage Posters') are fully visible & accessible.	P1	P / F / B
[ ]	TC-ADM-03	**Poster Banner Management (Upload & Edit)**	1. Open Poster Management modal. 2. Upload banner image. 3. Set Title, Badge, Subtext. 4. Save.	Poster saves to MongoDB Atlas 'banner_posters' collection and immediately becomes available for display.	P1	P / F / B
[ ]	TC-ADM-04	**Poster Action Button & Status Toggles**	1. Toggle Action Button OFF. 2. Toggle Status between Online and Offline. 3. Save and verify live.	Changes persist in MongoDB Atlas; customer carousel reflects button visibility and online status immediately.	P1	P / F / B
[ ]	TC-ADM-05	**Product CRUD & Stock Management**	1. In Admin, add new product with image and price. 2. Edit price/stock of existing item. 3. Delete test item.	Catalog updates in real time; stock changes immediately reflect in customer view.	P1	P / F / B
[ ]	TC-ADM-06	**Live Orders Dispatch Management**	1. Open Admin 'Track Bookings'. 2. Accept incoming order, change status to 'Out for Delivery' and 'Delivered'.	Status updates instantly in database; customer order tracking page reflects changes in real time.	P1	P / F / B
[ ]	TC-ADM-07	**Data Analytics & Revenue Dashboard**	1. Click 'Data Analytics' tab in Admin. 2. Check total revenue, total orders, top items.	Metrics match database orders; charts render cleanly without NaN or console errors.	P2	P / F / B

7. MOBILE RESPONSIVENESS, CROSS-BROWSER & PERFORMANCE

[X]	Test ID	Test Scenario / Feature	Steps to Execute & Test Inputs	Expected Acceptance Criteria	Pri	Status
[ ]	TC-PRF-01	**Mobile Viewport & Touch Optimization**	1. Test app on 375px (iPhone SE), 390px (iPhone 14), and 412px (Android) screens.	No horizontal scroll overflow; font sizes legible; touch targets >= 44px; hamburger menu opens smoothly.	P1	P / F / B
[ ]	TC-PRF-02	**Cross-Browser Compatibility**	1. Test core flows on Google Chrome, Mozilla Firefox, Apple Safari (iOS/Mac), and Microsoft Edge.	Visual styles, gradients, modals, sticky bars, and local state function consistently across all modern browsers.	P1	P / F / B
[ ]	TC-PRF-03	**Production Cloud Deployment Check (Render)**	1. Navigate to live Render URL (freshkart-9j9b.onrender.com). 2. Check browser console for errors.	App loads with valid HTTPS; API endpoints respond (<500ms); zero unhandled JavaScript or syntax errors in console.	P1	P / F / B
[ ]	TC-PRF-04	**Offline & Network Drop Resilience**	1. Throttle network to 'Offline' in DevTools Network tab while navigating.	Friendly offline toast notification shown; app gracefully recovers state once connection is restored.	P2	P / F / B
[ ]	TC-PRF-05	**Page Load Speed & Asset Compression**	1. Run Lighthouse audit on desktop and mobile view.	Lighthouse Performance score > 85; image assets (posters, icons) load within 1.5 seconds on 4G network.	P2	P / F / B

QA SIGNOFF & PRODUCTION RELEASE AUTHORIZATION

Total Tests Run: ____ / 42	Passed: ____	Failed: ____	Blocked: ____	Pass Rate (%): ____%
QA Lead Signature: _____	**Release Recommendation:** [ ] APPROVED FOR PRODUCTION [ ] REJECTED (BLOCKERS EXIST)	**Authorized By (Engineering Lead):** _____	**Sign-off Date:** ____ / ____ / 2026	**Target Build Commit:** Commit: 38491db (Clean)